

The AI talent market: A Statistical Analysis of Salary Gaps and Work Mode Trends

Overview

This project investigates the AI job market by analyzing the relationships between key variables. Specifically, it examines how salary correlates with company size and work modes, and whether company sizes and work modes are related to each other. A series of tests and complementary analyses were performed in order to examine these relationships. Primarily, the correlation between salary and considered extremely flexible and inflexible work modes (remote and on-site, respectively) correlation were examine, in which a non-significant relationship between them was found ¹. On the other hand, a strict correlation between company sizes and salaries was measured. In addition, the relationship examination between company sizes and work modes (this time considering all types of flexible and non-flexible modes) showed a strong correlation among them, forming a U-shaped pattern with startups (left peak) making extensive use of this work mode, mid-sized (bottom) companies restricting it as the scale, and large enterprises (right peak) reintroducing it using global infrastructure ³.

1. The test was conducted in order to evaluate the pay gap in salaries across extremely flexible and non-flexible work arrangements cultures.
2. Due to the imbalanced sample size across company tiers, a non-parametric Kruskal-Wallis test was applied, followed by a Mann-Whitney U post hoc test to draw specific salary gaps.
3. Although the test would prove the correlation between the variables, the relationship was not emphasized. Posteriorly, a heatmap was plotted in order to visualize this interaction.

Dataset

The project dataset contains 71,913 records of AI jobs between 2020 and 2026. Some cleaning approaches were applied, considering the tests that could be applied, in order to avoid biased results. The strategy consisted of verifying the test variables, finding outliers, and using an interquartile (IQR) technique to drop values falling beyond this range. In addition, another aspect

considered was the data density, specifically those related to categorical variables, since a significant spread difference between them could also generate a biased result. These features were considered to determine the most suitable test for each hypothesis in the generation of a valuable conclusion.

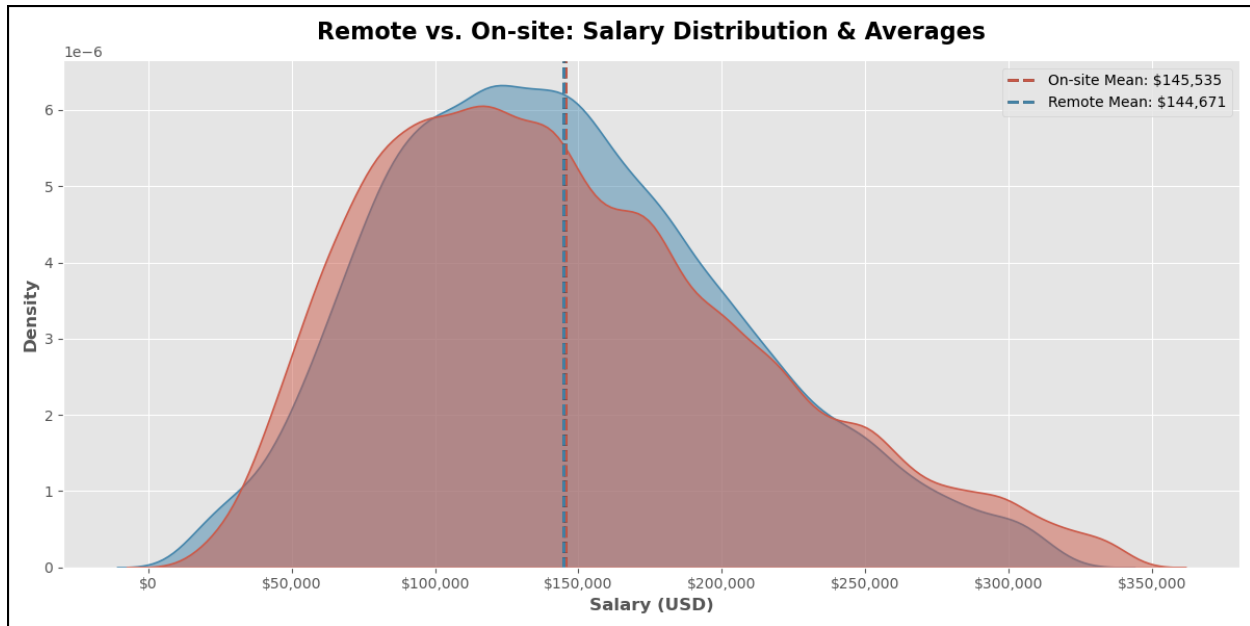
Findings

Experiment 1: Is there a salary gap between on-site and remote modes?

The first experiment was conducted to determine whether remote workers face a compensation penalty compared to those who work in traditional offices. Salary (USD) and work mode variables were set for the test performed in order to measure whether mean salaries are identical or not across work modes (on-site and remote). The test definitions and results are below:

- **Null Hypothesis (H_0):** $\mu_{\text{remote}} = \mu_{\text{onsite}}$ (mean salaries are identical between the work modes)
- **Alternative Hypothesis (H_1):** $\mu_{\text{remote}} \neq \mu_{\text{onsite}}$ (mean salaries are not identical between the work modes)
- **Alpha (α):** 0.05

To guarantee a precise result, Levene's test was performed, generating a p-value of 1.37×10^{-34} , and assuming that it is below a 0.05 significance threshold, consequently, Welch's t-test was selected to evaluate whether there is a significant salary difference between on-site and remote modes. A p-value of 0.11 was found, leading to the argument that there is no significant salary difference between these types.



The graphic above highlights the salary density of on-site and remote modes. Notice that both salary means are very close, which shows the similarity between their modes. Therefore, the obtained results from this experiment showed that remote and on-site modes are not generally strictly different in terms of salary range; accordingly, other aspects of the data set were evaluated to better comprehend the AI job market.

Experiment 2: Is there a significant salary difference across company sizes?

To deepen understanding of how salary is distributed across different company sizes, another experiment was conducted. Salary (USD) and company size (Large, medium, and small) variables were compared in order to evaluate this correlation. The test definitions and results are below:

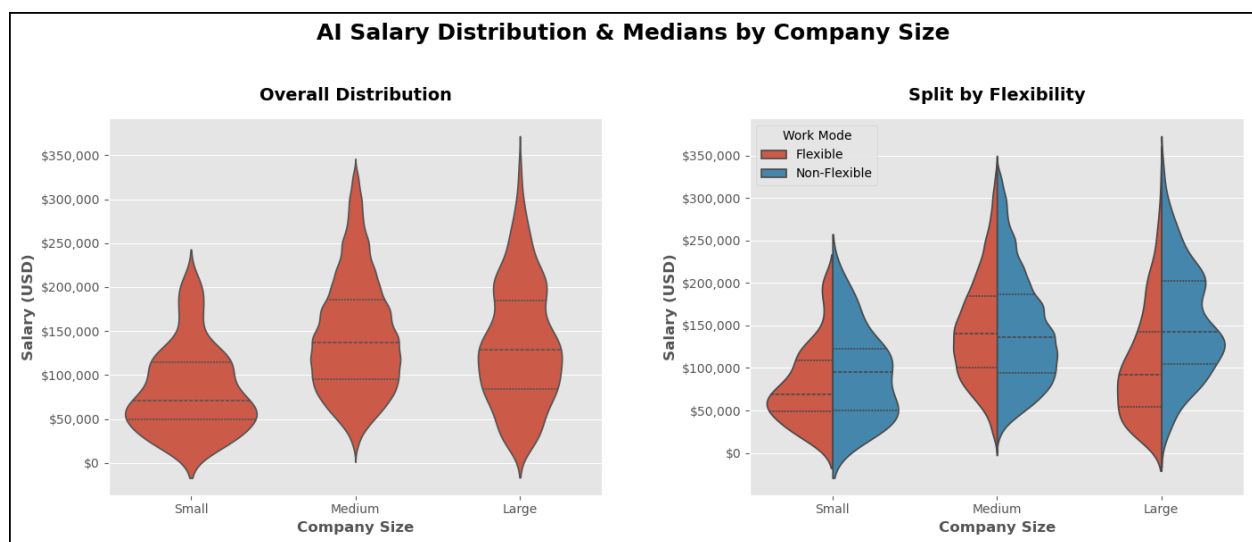
- **Null Hypothesis (H_0):** $\mu_S = \mu_M = \mu_L$ (Company size has no effect on average salary).
- **Alternative Hypothesis (H_1):** At least one company size has a different population mean salary.
- **Alpha (α):** 0.05

Firstly, the company size data spread was accounted for to check whether there is enough amount of data across different sizes. Once the company size data variation was strictly dissimilar and small-sized data was considerably less across companies, the Kruskal-Wallis test was selected

to continue examining this relationship. A p-value of 3.44×10^{-50} was found, highlighting that salary and company size are related.

Company Size	Count	Density (%)
Small	215	0.30%
Medium	1,588	2.21%
Large	70,110	97.49%
Total	71,913	100%

Subsequently, a Mann-Whitney U post hoc test was performed evaluating the correlation between all sizes and showed a strong correlation between medium and large companies, as well as between other sizes. In addition, a violin graph was plotted in order to better visualize the salary behavior across sizes, and later another allowing it to be broken down by flexible and non-flexible work modes.



The graphics above are an example of how additional information can modify perspective. When the median salary is analyzed alone, an almost linear progression is observed, where salary grows proportionally with company size, even though with a slight difference between medium and large companies. Afterwards, the split distributions reveal that the lower median salaries within both small and large companies are heavily skewed by flexible positions. Conversely, mid-sized companies maintain a much tighter salary grouping regardless of flexibility.

These findings indicate a U-shape pattern, which is formed as company size grows. Small companies are located at the left peak, indicating that they tend to adopt remote work as an

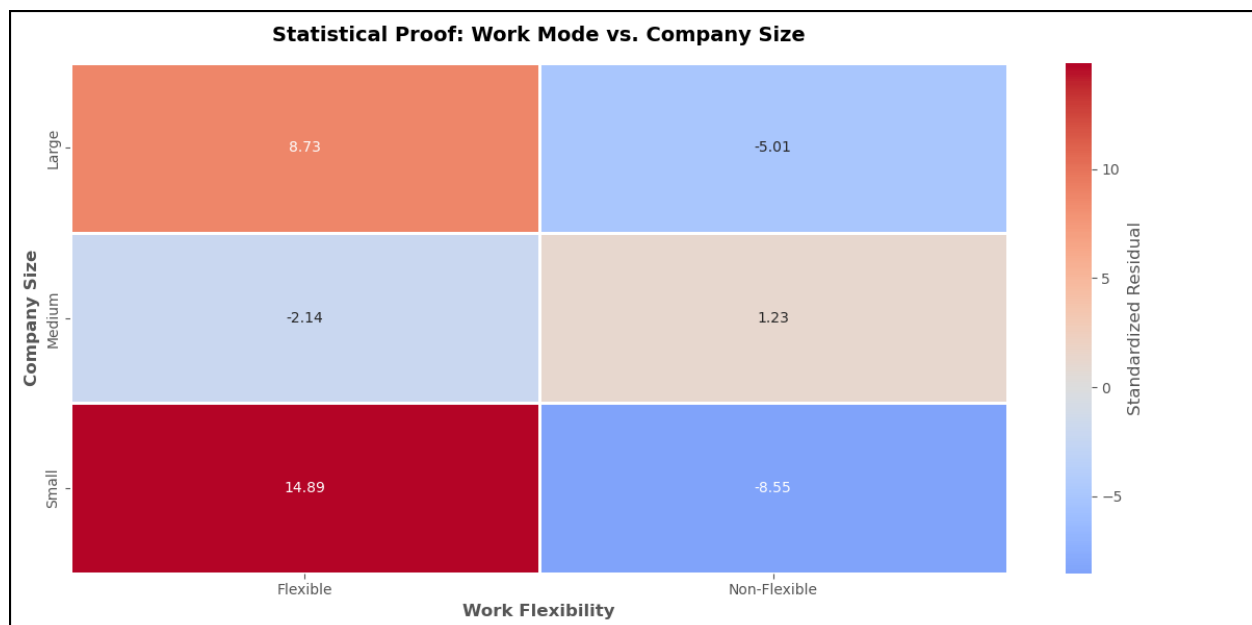
alternative to salaries that they cannot afford to take advantage of, as a strategy to face the offers of large companies. At the U bottom, medium companies' flexibility declines; they restrict remote work and enforce on-site policies. Even more, at the right peak are located large companies, indicating an exploitation of their superior budget and infrastructure to easily support globally distributed remote teams.

Experiment 3: Are different company sizes more related to flexibility work culture?

To evaluate the observed correlation between company sizes and work modes, another test was conducted. The test definitions and results are below:

- **Null Hypothesis (H_0):** Company size and work mode are completely independent variables
- **Alternative Hypothesis (H_1):** There is a statistically significant association between company sizes and work mode
- **Alpha (α):** 0.05

Once it had two categorical variables, a Chi-Square test was selected to measure this association. A p-value of 4.958×10^{-88} was found, leading to the argument that there is a statistically significant association between them. Although a test result would be enough to prove the correlation, a heatmap graphic was plotted in order to visualize it.



Although the discussed companies' strategies we've seen previously cannot be precisely proven, the visual behavior shows with accuracy the U-shape formed, where small and large companies operate within a more flexible environment, and medium companies are more restricted in a non-flexible environment. They are too large to operate like an agile startup, but perhaps lack the established, global remote work infrastructure of a massive enterprise, leading them to rely heavily on traditional on-site operations.

Conclusion

Although there is no significant pay gap penalty between the two extremes of flexible and non-flexible work arrangements, when deep analyses are conducted considering other features, a different conclusion is observed. Company size is an important factor that influences the salary range and the work mode that is applicable, so that combination is designed to form a U-shape, where small and large companies represent the flexibility peaks; even though the experiments indicate two different motivations for both, where the work arrangement is selected as a strategic compensation for smaller salaries to face large companies and as a reflection of the superior budget and infrastructure, respectively. Medium companies, on the other hand, are at the U bottom and deal with a situation where they are too big to operate agilely like a small company, and too small to operate with a large infrastructure company, which leads them to work on traditional on-site operations.